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Berdakh Abibullaev, Ph.D.

Curriculum Vitae

Research Area

Neural Engineering: Brain-Computer/Machine Interfaces; EEG/MEG/ECoG/fNIRS signal processing.

Machine Learning: Neural Networks and Deep Learning, Kernel based learning, Support vector machines.

Education

02/2010 **Ph.D**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area* : EEG time-series prediction, Machine learning and Neural Networks;

08/2006 **M.Sc**, *Yeungnam University*, Department of Electronic Engineering, South Korea.
- *Research area*: Wavelet Analysis, Time-frequency localization; Physiological feature extraction;

07/2004 **B.Sc**, *Tashkent University of Information Technologies*, Uzbekistan.

Academic Work Experience

09/2015-present **Assistant Professor**, *Robotics & Mechatronics Department, School of Engineering and Digital Science, Nazarbayev University*, Nur-Sultan, Kazakhstan.

Research Focus:
- Deep learning algorithms for Brain-Computer/Machine Interfaces.
- Brain-Computer/Machine Interfaces in Neural-Rehabilitation after Stroke.
- Transfer learning applications in session/subject independent Brain-Computer/Machine Interfaces.
- High-Level Robotic Manipulator Control via Brain-Computer Interfaces for Assistive Applications

5/2018 - 7/2018 **Visiting Professor**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- Cortical source imaging of electroencephalographic signals in patients with stroke to examine neural correlates brain plasticity pre and post clinical trials with BMI. actuated exoskeleton.

5/2014 - 9/2015 **NIH Postdoctoral Research Fellow II**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.
- BMI-Exo Project: Brain-Machine Interface Control of a Therapeutic Exoskeleton for Neurorehabilitation.
  - Estimating neural correlates and mechanisms of an external command (mental intent) used to drive the external therapeutic exoskeleton for BMI.
- Neural Correlates of Infant Mirror Neuron System
  - Investigate the role the mirror neuron system plays in the motor learning of young infants via noninvasively acquired brain signals;
  - Categorizations of large-scale brain connectivity analysis to understand neural mechanisms of infant behavior.
- Electromagnetic-source localization and estimating brain network in connectivity analysis

01/2014-05/2014 **Research Scientist**, *Department of Neurology, Samsung Medical Center*, Seoul, South Korea.
- Investigated High-Frequency Oscillations of ECoG/EEG signals relevant to Epileptogenesis using machine learning algorithms. Supported medical doctors through brain data analysis in the presurgical evaluation of patients with refractory epilepsy.
- Neuroelectromagnetic source localization from scalp EEG & MEG recordings for delineation of epileptogenic zones in the surgical treatment of refractory focal epilepsy.

08/2013-12/2013 **Assistant Professor**, *Robotics & Mechatronics Department, School of Science and Technology, Nazarbayev University*, Astana, Kazakhstan.

Teaching: Enhancing Brain-Computer Interfaces by Machine Learning Techniques :
- Signals and Sensing with Laboratory (*a sophomore-level course*).

04/2011-08/2013 **Senior Research Scientist**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

Development of rehabilitation strategies for stroke patients via robot-assisted physical therapy with BCIs.
- Implemented kernel-based machine learning algorithms for online BCIs to control neuro-rehabilitation devices.
- Examined inter-subject/session variabilities to enhance fNIRS-based BCI performances.
- Design and optimization of non-invasive based BCI paradigms for real-time robot control.

01/2010-04/2011 **Postdoctoral Research Scholar**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

Brain-Computer Interfaces in therapy and diagnosis of Attention Deficit and Hyperactivity Disorders (ADHD).

- Involved in quantitative EEG and Neurofeedback methods of ADHD children. Time-frequency localization; Identification of strongly relevant neural signal patterns of ADHD using machine learning.
- Developed machine learning-based semi-supervised feature selection and classification algorithms.
- Participated in the developmental assessment of ADHD children during a long-term neuro-therapy in cooperation with a neurorehabilitation center at Daegu University.

Honors and Awards

11/2017 Best paper award: *fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network*," in Multisensor Fusion and Integration for Intelligent Systems, 2017 IEEE International Conference.

06/2014 Best poster award on "*The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis*," Korean Epilepsy Congress 2014.

12/2012 Best Researcher of the Year Award. Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology, Daegu, South Korea.

01/2011 Enlisted in Marquis Who's Who in the World 2011, 28th Edition, USA

09/2006 Korean Government Information Technology Scholarship for International Graduate Students for the Ph.D. program, South Korea.

09/2004 Brain Korea #21 Scholarship for the MSc program.

Technical Skills

Software skills.

Expert in the Python language and libraries such as Cython; very good knowledge of C, C++. Experience with various tools and languages, including bash, LATEX, HTML, Git, and different database query languages.

Data Science.

Expert knowledge in NumPy, Scipy, Pandas, Sklearn, PyTorch Deep larning libraries

Neural data acquisition systems.

Expertise with Nicolet EEG Amplifiers, Blackrock Microsystems (intracranial recording), Functional Near-Infrared Spectroscopy FOIRE3000, BIOPAC fNIR200 Optical Brain Imaging, BioSemi Active-Two EEG system, Brain Products EEG amp., BIOPAC MP100 EEG amp. And G.tec g.USBamp systems.

Productivity applications.

TEX: [LATEX, BibTEX, PSTricks, Beamer].

System Environments.

Linux (Pop_OS, Ubuntu), Windows

Service to the Scientific Community

Associate Editor.

- IEEE Access Journal (Impact Factor: 3.7)
- PeerJ Computer Science (Impact Factor: 3.09)

Occasional Reviewer.

- IEEE Access Journal
- Neuroimage, Elsevier.
- IEEE Transactions on Neural Networks and Learning Systems
- Brain Research, Elsevier.

Program Committee Member and Session Chair.

- International Workshop on Artificial Intelligence for Healthcare Applications at International Conference on Pattern Recognition 2020.
- Session Chair: The 6th International Conference on Brain-Computer Interface (BCI), IEEE, 2018, South Korea.
- Session Chair: IEEE International Conference on Multisensory Fusion and Integration for Intelligent Systems, 2017, Daegu, South Korea.
- Session Chair: International Conference on Control Automation and Systems Oct.27-30, 2010, Seoul, South Korea

Project Proposals, Participation & Coordination

01/2021 **Principal Investigator**, ***Brain-Machine Interfaces for Neural Rehabilitation after Stroke***, This project aims to accelerate the efficacy of robotic rehabilitation using a new neural interface system developed for upper-limb repair after stroke. It aims to facilitate expediate motor recovery from incomplete upper-limb paralysis and investigate in-depth understanding of cortical reorganization in the brain associated with the recovery..

01/2021 **Co-Investigator**, ***Classifier Validity and Its Applications in Bioinformatics***, The project studies the measures of classifier validity and explores their applications in bioinformatics. Nazarbayev University..

01/2019 **Co-Investigator**, ***Neural-assisted optimization of human learning with multimodal neuroimaging of EEG and NIRS***, Implementing and validate a neural-assisted, machine-learning-based intelligent tutoring system that uses information from physiological signals to adapt the learning schedule to a user's individual learning capacity..

09/2015 **Principal Investigator**, ***Development of a Brain-Computer Interface with a Robot and Functional-Electrical Stimulation for Upper-Limb Stroke Rehabilitation (BCI-ROBOFES)***, Evaluated by the committee of Oak Ridge Associated Universities (ORAU);.

9/2014 **Participant**, *Cognitive-Motor Development Research in Infants.*, Lead a team of Ph.D. students in exploring & decoding of the neural basis of *Imitation and Action Understanding* in freely-behaving infants via multimodal brain imaging techniques, University of Houston, Texas.

9/2014 **Participant**, ***The BMI-Exo Project: Facilitating Personalized Robotic Rehabilitation Of The Upper Limb, [NIH Grant R01NS081854-02, National Robotics Initiative]***, Lead a part of research that focuses on understanding neurophysiological correlates of motor learning during movements in stroke patients..

04/2014 **Participant**, ***Development of high-resolution subdural electrodes and data mining methods for precise localization of epileptogenic zone in treatment of patients with refractory epilepsy.***, Participated in idea generation and in drafting/writing of the project proposal at Samsung Medical Center, Korea.

Publications

Patents

[1] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Rehabilitation Training System and Method**. *U.S. Patent No*: 9,081,890,Washington, DC: United States. Patent and Trademark Office, 2015 (https://www.google.com/patents/US9081890).

[2] J. An, S.H. Jin, S.H. Lee, J.I. Moon B. Abibullaev, J.H. Ahn and G.H. Jang. **Self-directed rehabilitation training method combining brain signals and functional electro-stimulation**. *EFS ID*: 17077057, 09-OCT-2013, United States, (http://www.google.com/patents/US20140200632).

[3] J. An, S.H. Jin, S.H. Lee, B. Abibullaev, J.I. Moon , J.H. Ahn and G.H. Jang. **Combining Brain Signals and Functional Electrostimulation Self-Directed Rehabilitation Method**, Patent Registration Number: 101501524000, 05.03.2015. South Korea,(http://eng.kipris.or.kr/).

Refereed Journal Publications

[1] B. Abibullaev and A. Zollanvari, **A Systematic Deep Learning Model Selection for P300-Based Brain-Computer Interfaces**, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, DOI: 10.1109/TSMC.2021.3051136, 2021.(https://ieeexplore.ieee.org/document/9351887)***(Impact Factor: 13.451**; Quartile rank: Q1)*.

[2] A. Zollanvari and B. Abibullaev **Bias Correction for Linear Discriminant Analysis**, *Pattern Recognition Letters*, vol. 151, pp. 41-47, 2021, https://doi.org/10.1016/j.patrec.2021.07.026 *(Impact Factor: 3.7; Quartile rank: Q1)*.

[3] B. Abibullaev and A. Zollanvari, **Learning Discriminative Spatiospectral Features of ERPs for Accurate Brain-Computer Interfaces** *IEEE Journal of Biomedical and Health Informatics*, vol. 98, pp. 1-12, 2019, DOI: 10.1109/JBHI.2018.2883458 *(Impact Factor: 5.772; Quartile rank: Q1)*.

[4] B. Abibullaev, I. Dolzhikova and A. Zollanvari, **A Brute-force CNN Model Selection for Accurate Classification of Sensorimotor Rhythms in BCIs** *IEEE Access*, 2020, DOI: 10.1109/ACCESS.2020.2997681 *(Impact Factor: 3.36; Quartile rank: Q1)*.

[5] B. Abibullaev, A. Zollanvari, B. Saduanov and T. Alizadeh, **Design and Optimization of a BCI-Driven Telepresence Robot Through Programming by Demonstration** *IEEE Access*, August 2019, DOI: 10.1109/ACCESS.2019.2933268 *(Impact Factor: 3.36; Quartile rank: Q1)*.

[6] A. Zollanvari, M. Abdirash, A. Dadlani and B Abibullaev. **Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis for Cost-Sensitive Binary Classification** *IEEE Signal Processing Letters*, 22 May 2019, DOI: 10.1109/LSP.2019.2918485 *(Impact Factor: 3.1; Quartile rank: Q1)*.

[7] B. Abibullaev, J An, SH Lee, JI Moon. **Design and Evaluation of Action Observation and Motor Imagery based BCIs using NIRS** *Measurement*,vol. 98, pp. 250-261, 2017, Elsevier *(Impact Factor: 3.927; Quartile rank: Q1)*.

[8] N.A. Bhagat, A. Venkatakrishnan, B. Abibullaev, E.J. Artz, N. Yozbatiran, A. Blank, J. French, C. Karmonik, R.G.Grossman, M.K O'Malley, G. Francisco, J.L. Contreras-Vidal. **Design and optimization of an EEG-based brain-machine interface (BMI) to an upper-limb exoskeleton for stroke survivors.** *Frontiers in Neuroscience*, vol. 10, March, 2016 (Impact Factor: 4.677; Quartile rank: Q1).

[9] J.G. Cruz-Garza, Z.R. Hernandez, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **A novel experimental and analytical approach to the multimodal neural decoding of intent during social interaction in freely-behaving human infants.** *Journal of Visualized Experiments*, DOI: 10.3791/53406, October, 2015 (Impact Factor : 1.5; Quartile rank: Q2).

[10] C.H. Park, J.H Seo, D. Kim, B. Abibullaev, H. Kwon, Y.H. Lee, M.Y. Kim, K. Kim, J.S. Kim, E.Y. Joo, S.B. Hong, (2015, Feb). **EEG Source Imaging in Partial Epilepsy in Comparison with Presurgical Evaluation and Magnetoencephalography.** *Journal of Clinical Neurology*, 2015 Feb 17, 11:e12 (Impact Factor : 3.077; Quartile rank: Q2).

[11] B. Abibullaev, J An, S.H. Jin, and J.I. Moon. **Classification of brain hemodynamic signals arising from visual action observation tasks for brain-computer interfaces : An fNIRS study.** *Measurement*, DOI:10.1016/j.measurement.2013.11.044, 2014. Elsevier (Impact Factor : 3.927; Quartile rank: Q1).

[12] B. Abibullaev, J An, S.H. Lee, S.H. Jin, and J.I. Moon. **Minimizing inter-subject variability in fNIRS based brain computer interfaces via multiple-kernel support vector learning.** DOI: 10.1016/j.medengphy.2013.08.009, *Medical Engineering Physics*, 2013. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[13] B. Abibullaev and J. An. **Classification of frontal cortex hemodynamic response during cognitive tasks using wavelet transforms and machine learning algorithms.** *Medical Engineering Physics*, 34(10):1394–410, 2012. Elsevier (Impact Factor : 2.242; Quartile rank: Q2).

[14] B. Abibullaev and J. An. **Decision support algorithm for diagnosis of ADHD disorder using electroencephalograms.** *Journal of Medical Systems*, 36(4):2675–2688, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[15] B. Abibullaev, J. An, and J.I. Moon. **Neural network classification of brain hemodynamic responses from four mental tasks.** *Int J Optomechatronics*, 5(4):340–359, 2011. Taylor & Francis (Impact Factor : 1.9; Quartile rank: Q2).

[16] B. Abibullaev and H.D. Seo. **A new QRS detection method using wavelets and artificial neural networks.** *Journal of Medical Systems*, 35(4):683–691, 2011. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[17] B. Abibullaev, M.S. Kim, and H.D. Seo. **Epileptic spike detection using continuous wavelet transforms and artificial neural networks.** *Int J of Wavelets, Multires and Inform Proc*, 8(1):33–48, 2010. Worldscientific (Impact Factor : 0.463; Quartile rank: Q3).

[18] B. Abibullaev, M.S. Kim, and H.D. Seo. **Seizure detection in temporal lobe epileptic EEGs using the best basis wavelet functions.** *Journal of Medical Systems*, 34(4):755–765, 2010. Springer (Impact Factor : 4.460; Quartile rank: Q2).

[19] M.S. Kim, Y.C. Cho, B. Abibullaev, and H.D. Seo. **Analysis of brain functions and classification of sleep stage EEG using daubechies wavelet.** *Sensors and Materials*, 20(1):1–15, 2008. MYU, Japan (Impact Factor: 0.92; Quartile rank: Q3).

Conference proceedings

[1] I. Dolzhikova, B. Abibullaev, R. Sameni, A. Zollanvari, "**An Ensemble CNN for Subject-Independent Classification of Motor Imagery-based EEG**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[2] Y. Kainolda, B. Abibullaev, R. Sameni, A. Zollanvari, "**Is Riemannian Geometry Better than Euclidean in Averaging Covariance Matrices for CSP-based Subject-Independent Classification of Motor Imagery?**", the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBS), October 31 – November 4, 2021

[3] M. Nurpeiissov, B. Abibullaev, T. Alizadeh, "**A novel human-robot interaction framework based on Telegram and programming by demonstration**", 9th International Conference on Robot Intelligence Technology and Applications (RITA 2021), on Dec 16 - 17, 2021 at KAIST Daejeon, Korea.

[4] S. Abilkassov, M. Kairgaliyev, B. Zhakanov, B. Abibullaev, "**A System For Drivers' Cognitive Load Estimation Based On Deep Convolutional Neural Networks and Facial Feature Analysis**", in 22nd IEEE International Conference on Industrial Technology, Virtual 10-12, 2021, Spain.

[5] K. Kunanbayev, B. Abibullaev, A. Zollanvari, "**Data Augmentation for P300-based Brain-Computer Interfaces Using Generative Adversarial Networks**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-7, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385317.

[6] K. Kunanbayev, D. Azhigulov, B. Abibullaev, A. Zollanvari, "**Deep Transfer Learning for Subject-Independent ERP-based BCIs**", in 2021 9th International Winter Conference on Brain-Computer Interface (BCI), 22-24 Feb. 2021, pp. 1-3, Gangwon, Korea (South), doi: 10.1109/BCI51272.2021.9385311.

[7] A. Oleinikov, B. Abibullaev, M. Folgheraiter, "**On the Classification of Electromyography Signals to Control a Four Degree-Of-Freedom Prosthetic Device**," in 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), July 20-24, Montreal, Canada.

[8] B. Saduanov, D. Tokmurzina, K. Kunanbayev and B. Abibullaev, **"Design and Optimization of a Real-Time Asynchronous BCI Control Strategy for Robotic Manipulator Assistance"**, in 2020 8th International Winter Conference on Brain-Computer Interface. (BCI), 2020, GangWon, South Korea.

[9] Adilet Tuleuov and B. Abibullaev, "**Deep Learning Models for Subject-Independent ERP-based Brain-Computer Interfaces**," in *the 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*, March 20-23, 2019, San Francisco, CA, USA.

[10] B. Abibullaev, Y. Orazayev, and A. Zollanvari, "**Novel Spatiospectral Features of ERPs Enhances Brain-Computer Interfaces**," in *Brain-Computer Interface (BCI), 2019 7th International Conference on*. IEEE Brain Society, 2019, GangWon, South Korea.

[11] B. Saduanov, D. Tokmurzina, T. Alizadeh, and B. Abibullaev, "**Brain-computer interface humanoid pre-trained for interaction with people**," in *2018 ACM/IEEE International Conference on Human-Robot Interaction*. ACM, March 5-8, 2018, pp. 229–230, Chicago, IL, USA.

[12] A. Oleinikov, B. Abibullaev, A. Shintemirov, and M. Folgheraiter, "**Feature extraction and real-time recognition of hand motion intentions from emgs via artificial neural networks**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1-5, GangWon, South Korea.

[13] G. Lee, S. H. Jin, S. T. Yang, J. An, and B. Abibullaev, **"Cross-correlation between HBO and HbR as an effective feature of motion artifact in the fNIRS signal,"** 2018 6th International Conference on Brain-Computer Interface (BCI), pp. 1-3, 2018, IEEE Brain Society, GangWon, South Korea.

[14] B. Saduanov, T. Alizadeh, J. An, and B. Abibullaev, "**Trained by demonstration humanoid robot controlled via a BCI system for telepresence**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE Brain Society, 2018, pp. 1–4, January, GangWon, South Korea.

[15] G. Lee, S. H. Jin, S. H. Lee, B. Abibullaev, and J. An, "**fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network**," in *Multisensor Fusion and Integration for Intelligent Systems (M.F.I.), 2017 IEEE International Conference on*. IEEE, 2017, pp. 186–193, Daegu, South Korea.

[16] A. Zhumadilova, D. Tokmurzina, A. Kuderbekov, and B. Abibullaev. **Design and Evaluation of a P300 Visual Brain-Computer Interface Speller in Cyrillic Characters**. In *the 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), IEEE*, 2017, Aug 28 - Sept 1, Lisbon, Portugal.

[17] B. Abibullaev. **Learning Suite of Kernel Feature Spaces Enhances SMR-Based EEG-BCI Classification.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, GangWon, South Korea.

[18] D. Nurseitov, A. Serekov, A. Shintemirov and B. Abibullaev. **Design and Evaluation of a P300-ERP based BCI System for Real-Time Control of a Mobile Robot.** In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, Gangwon , South Korea.

[19] B. Abibullaev and J An. **On Robust Classification of Hemodynamic Signals for BCIs via Multiple Kernel ν-SVMs.** *Intelligent Robots and Systems (IROS), 2016 IEEE/RSJ International Conference*, 2016, Oct10-15, Daejeon, South Korea.

[20] J.G. Cruz-Garza, Z.R. Hernandez, M. Megjhani, B. Abibullaev, T. Tse, E. Caducoy, and JL Contreras-Vidal. **Neural Development of social cognition in the first two years of life - Early findings.** In *Society for Neuroscience* 2015 Oct 17-21; Chicago, USA.

[21] Z.R. Hernandez, A.J. Arenas-Castellanos, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **Decoding Intent From Non-invasive EEG in Freely Behaving Infants.** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[22] A.J. Arenas-Castellanos, Z.R. Hernandez, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal **A developmental Analysis of Behaviors Related to the Mirror Neuron System in 6-24 Months Infants** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, U.S.A.

[23] Z.R. Hernandez, J.G. Cruz-Garza, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **Supervised Classification of Intended Behaviors Using Electroencephalography (EEG) from Freely-Behaving Infants: Early Findings.** In *12th Annual G.C.C. Conference on Theoretical and Computational Neuroscience*, Rice University, Houston, TX, United States, February 6-7, 2015.

[24] N. A. Bhagat, A Venkatakrishnan, B. Abibullaev, E. J. Artz, A.A. Blank, J. A. French, N. Yozbatiran, and R. G. Grossman, M. K. O'Malley, J. L. Contreras-Vidal, G. E. Francisco **BMI Control of a Therapeutic Exoskeleton to Facilitate Personalized Robotic Rehabilitation of the Upper Limb.** *In National Robotics Initiative (N.R.I.) P.I. Meeting*, Westin Arlington, Nov. 19-20 2014, United States.

[25] C.H.Park, B. Abibullaev, E.Y. Joo, S.C. Hong and S.B. Hong. **The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis.** In *the 19th Korean Epilepsy Congress*, Seoul, Republic of Korea, June 12-14 2014.

[26] S.H. Lee, J. An, G. Jang, S.H. Jin, B. Abibullaev, and J.I. Moon. **Neural activity during observation, imagery, and execution of eating: An fNIRS pilot study.** In *the 19th Annual Meeting of the Organization for Human Brain Mapping*, Seattle, WA, United States, June 16-20, 2013.

[27] J. An, S.H. Jin, S.H. Lee, G. Jang, B. Abibullaev, J. Ahn, H. Lee, and J.I. Moon. **Cortical activation pattern for grasping during observation, imagery, execution, FES, and observation-FES integrated BCI: An fNIRS pilot study.** In *the 35th Annual Int. Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Osaka, Japan, July 3-7 2013.

[28] J. An, S.H. Lee, S.H. Jin, B. Abibullaev, G. Jang, J. Ahn, H. Lee, and J.I. Moon. **The beginning of neurohaptics: Controlling cognitive interaction via brain haptic interface.** In *the 2013 IEEE International Winter Workshop on Brain-Computer Interface*, Gangwon Province, Korea, Feb. 18-20 2013.

[29] B. Abibullaev, J. An, J.I. Moon, S.H. Lee, and S.H. Jin. **A study on the BCI-Robot assisted stroke rehabilitation framework using brain hemodynamic signals.** In *the 9th Int Conference on Ubiquitous Robots and Ambient Intelligence*, IEEE proceedings, Daejon, Korea, November 26-29 2012.

[30] B. Abibullaev, J. An, S.H.Lee, S.H. Jin, J.I Moon, H.J. Lee. **Decoding of Hemodynamic Brain Responses for Brain-Computer Interfaces via Ensemble Support Vector Learning.** In *the Human-Computer Interaction Conference*, Gangwon-Do, Korea, January 11-13, 2012.

[31] B. Abibullaev, W.S. Kang, S.H. Lee, J. An, and H.D. Seo. **Near-Infrared Spectroscopy in the analysis of functional brain activity during cognitive tasks.** In *2010 IEEE Sensors Conference*, Hawaii, United States, November 1-4, 2010.

[32] B. Abibullaev, W.S. Kang, S.H. Lee, and J. An. **Recognition of brain hemodynamic mental response for brain-computer interfaces.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

[33] S.H. Lee, B. Abibullaev, W.S. Kang, and J. An. **analysis of attention deficit hyperactivity disorder in EEG using wavelet transform and self-organizing maps.** In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

[34] W.S. Kang, B. Abibullaev, S.H. Lee, and J. An. **Path planning algorithm using the values clustered by K-means.** In *the 15th Int. Symposium on Artificial Life and Robotics*, Japan, February 4-6, 2010.

[35] B. Abibullaev, H.D. Seo, W.S. Kang, and J. An. **A wavelet-based method for detecting and localizing epileptic neural spikes in EEG signals.** In *the 2nd Int. Conf. on Interaction Sciences: Information Technology, Culture, and Human. A.C.M.*, Seoul, Korea, Nov. 24- 26, 2009.

[36] B. Abibullaev and H.D. Seo. **Epileptic seizures detection using continuous time wavelet based neural networks.** In *the 6th International Conference on Information Technology: New Generations, IEEE Computer Society*, Las Vegas, Nevada, United States, April 27-29 2009.